Computer & IT / Secondary

Trace a simple algorithm

A trace table records how values change as an algorithm runs. Follow the instructions in order.

Core ideas

1. Create a column for each changing variable.
2. Record the initial values before running the loop.
3. Update the table after each instruction or iteration. Check the loop boundary carefully.

Worked example

Set total = 0. For n = 1, 2, 3, add n to total. The running total is 1, then 3, then 6. The final value of total is 6.

Try it yourself

1. What is total after adding 1, 2, 3 and 4 to an initial total of 0?
2. What does an initial value tell you?
3. Why check the loop boundary?

Check your answers

1. 10
2. The variable's value before the relevant instructions begin.
3. It determines which iterations run, including whether the final value is included.